

Business Requirements Documentation

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Business Problem

Northstar Insurance currently has no systematic way to identify policyholders at risk of not renewing before their renewal date arrives. Retention efforts today are reactive or generic, with no segmentation by risk level and no visibility into how much premium is exposed. Based on historical renewal data, approximately 9.1% of policyholders lapse at renewal, with no targeted outreach in place to address it.

(Northstar Insurance is a fictional mid-size Canadian P&C insurer created to contextualize this project. All data is synthetic.)

Objectives

1. Identify and segment policyholders by lapse risk using a transparent, explainable scoring model.
2. Quantify premium exposure by risk tier to prioritize retention effort by business impact.
3. Enable targeted outreach to high-risk policyholders through their preferred language and communication channel.

Project Scope

In Scope

- Policyholder data cleaning and validation
- Risk based segmentation of current policyholder book
- Business rule validation against historical outcomes
- Retention analytics and reporting (dashboard)
- Automated drafting of personalized outreach content
- Human approval workflow prior to outreach delivery

Out of Scope

- Machine learning based churn prediction
- Real time or event driven scoring
- Pricing or discount decisioning
- CRM or call centre system integration
- Unsupervised (non human reviewed) outreach delivery

Stakeholders

Stakeholder	Role in Project	Interest / Stake	Engagement
Retention Manager	Business sponsor, rule owner	Owns lapse rate target; defines what "high risk" means; accountable for program ROI	Accountable
Retention Agents	Primary users	Need a trustworthy, prioritized outreach queue with ready-to-review drafts	Responsible (day-to-day)
Compliance Officer	Governance	CASL/PIPEDA adherence; explainable decisions; human approval before any send	Consulted
Actuarial / Underwriting	Domain validation	Scoring logic must align with how the business already assesses risk	Consulted
IT / Data Services	Platform & data owners	Own source systems (policy, billing, claims) and the pipeline the ETL runs on	Informed / Supporting

Benefits

1. **Visibility into retention risk.** Northstar gains the ability to quantify how much premium is exposed to lapse, by risk tier, at any point in time.

Measured by: premium exposure available by tier and renewal window on the retention dashboard.

2. **Prioritized retention effort.** Retention agents stop treating all renewals equally and concentrate effort on customers with the highest lapse risk and premium value.

Measured by: share of high-risk policyholders contacted before their renewal date (contact rate).

3. **Efficient, consistent outreach.** Agents spend less time researching and drafting messages; outreach is generated in the customer's preferred channel and language and reviewed before sending.

Measured by: agent time per outreach, and 100% of sent messages passing through human approval.

4. **Reduced lapse and protected revenue.** Fewer high-risk customers lapse at renewal, reducing lost premium.

Measured by: lapse rate of contacted high-risk customers versus a comparable uncontacted holdout group.

Requirements

Functional Requirements

ID	Requirement	Priority
FR-01	The system shall ingest policyholder and renewal history data into a staging layer on a scheduled basis.	Must
FR-02	The system shall remove duplicate customer records, retaining the earliest occurrence, and log the count removed.	Must
FR-03	The system shall standardize inconsistent categorical values (e.g. contact channel casing) against a defined reference format.	Must
FR-04	The system shall quarantine records that fail validation (e.g. renewal date preceding policy start date) into an exceptions table rather than deleting them.	Must
FR-05	The system shall calculate a risk score for every active policyholder using the defined business rules and assign a risk tier.	Must
FR-06	The system shall display policyholder counts, premium exposure, and priority outreach windows by risk tier on a dashboard.	Must

FR-07	The system shall generate a personalized outreach draft for each high-risk policyholder renewing within 90 days, using their preferred channel and language.	Must
FR-08	The system shall route every generated draft to a human reviewer for approval before any message is sent.	Must

Non - Functional Requirements

ID	Requirement	Priority
NFR-01	Every risk score shall be explainable as the sum of individually named business rules	Must
NFR-02	Customer data handling shall comply with PIPEDA; personally identifiable information shall be restricted to authorized roles.	Must

User Stories

ID	User Story	Acceptance Criteria	Satisfies
US-01	As a Retention Manager, I want to see premium exposure by risk tier, so that I can prioritize retention effort by business impact.	AC1: Dashboard displays total premium and count for High, Medium, and Low tiers. AC2: Figures update when the underlying data is rescored.	FR-06
US-02	As a Retention Agent, I want a prioritized list of high-risk customers renewing so that I focus my limited time on the customers most likely to lapse and worth the most premium.	AC1: The list contains only customers in the High risk tier. AC2: The list is sorted by risk score descending, with earliest renewal date breaking ties.	FR-06, FR-07
US-03	As a Retention Agent, I want a personalized outreach draft in the customer's preferred channel and language, so that I can review and send it quickly without writing it myself.	AC1: The draft matches the customer's preferred channel and language, formatted appropriately (e.g., under 300 characters for Text, subject plus body for Email) AC2: The draft never	FR-07

		references risk scores, missed payments, or claim status.	
US-04	As a Compliance Officer, I want every outreach draft to be approved by a human, so that no message reaches a customer without review.	AC1: Every draft enters a "Pending Review" status and cannot be sent until approved. AC2: Every approval or rejection decision is recorded with a timestamp.	FR-08